Cassidy K. Buhler, Ph.D.

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Professional Appointments

2024 – Present **Postdoctoral Associate** Boulder, CO
Environmental Data Science Innovation & Impact Lab (ESIIL)
University of Colorado, Boulder

Education

2024 **Ph.D. Operations Research** Philadelphia, PA
Computational Data Science Minor
Drexel University
Dissertation: Advances in Optimization with Applications to Biodiversity Conservation

2019 **B.S. Mathematics** Salt Lake City, UT
Statistics Emphasis
University of Utah

Publications

Peer-Reviewed Journal Articles

C. K. Buhler, H. Y. Benson, and D. F. Shanno, "Regularized step directions in nonlinear conjugate gradient methods," *Mathematical Programming Computation*, vol. 16, pp. 629–664, 2024, ISSN: 1867-2957. DOI: 10.1007/s12532-024-00265-9.

C. K. Buhler, R. S. Terry, K. G. Link, and F. R. Adler, "Do mechanisms matter? Comparing cancer treatment strategies across mathematical models and outcome objectives," *Mathematical Biosciences and Engineering*, vol. 18, no. 5, pp. 6305–6327, 2021, ISSN: 1551-0018. DOI: 10.3934/mbe.2021315.

Refereed Conference Proceedings

C. K. Buhler and H. Y. Benson, "Decision-making for land conservation: A derivative-free optimization framework with nonlinear inputs," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024, pp. 21 932–21 939. DOI: 10.1609/aaai.v38i20.30195.

C. K. Buhler and H. Y. Benson, "Optimal land conservation decisions for multiple species," in *Proceedings of the 52nd Northeast Decision Science Institute Annual Conference*, vol. 52, Washington, D.C., 2023, pp. 808–816.

Policy & Technical Reports

J. Howard, C. Schloss, R. Olliff Yang, A. Canales, C. K. Stanley, M. Webb, D. Ackerly, C. Boettiger, **C. Buhler**, S. Monfort, G. Bucciarelli, R. Meyer, A. Levi Simons, J. Bando, L. Chamberlin, S. Veloz, C. Dickison, A. Hill, R. Johnson, R. Hill, A. Kochanek, D. Rademacher, L. Micheli, K. Penrod, G. Rapacciuolo, N. Roth, D. Spatz, D. Gluesenkamp, and collaborators, "California's 30 x 30 initiative at its midpoint: A biodiversity assessment snapshot of the state's conservation areas," *Artemisia*, vol. 51, no. 2, Jan. 2026, California Native Plant Society. URL: https://www.cnps.org/artemisia-journal/californias-30-x-30-initiative-at-its-midpoint-44874.

Software

California 30x30 Planning & Assessment Tool (California Biodiversity Network Edition)
https://huggingface.co/spaces/boettiger-lab/ca-30x30-cbn
10.5281/zenodo.16988638 (2025)

LLM Decision-Support Prototype using Trust for Public Land Data
https://huggingface.co/spaces/boettiger-lab/tpl
10.5281/zenodo.18500742 (2026)

LandVote LLM Decision-Support Prototype
https://huggingface.co/spaces/boettiger-lab/landvote
10.5281/zenodo.18500795 (2026)

California 30x30 Planning & Assessment Prototype
https://huggingface.co/spaces/boettiger-lab/ca-30x30
10.5281/zenodo.14933818 (2025)

Derivative-Free Optimization for Land Conservation
https://github.com/cassiebuhler/conservation-dfo
10.5281/zenodo.13742960 (2024)

Conmin-CG: Hybrid Cubic Regularization of Conjugate Gradient Methods
https://github.com/cassiebuhler/ConminCG
10.5281/zenodo.13315592 (2024)

Datasets

State-level 30×30 conservation policy implementation and progress across the United States
https://https://source.coop/cassiebuhler/30x30-state-policy
10.5281/zenodo.18500907 (2026)

Working Groups

2026 – Present **Developing Best Practices for Large Language Models in Environmental Science**
Environmental Data Science Innovation & Impact Lab (ESIIL) Working Group

2025 **Maka Sitomniya: Preserving Mother Earth by Asserting Lakota Sovereignty in Earth Data Science**
Environmental Data Science Innovation & Impact Lab (ESIIL) Working Group

2024 – Present **California 30x30 Biodiversity Assessment**
California Biodiversity Network (CBN) Working Group

Fellowships & Research Experience

2024 – Present **Postdoctoral Fellowship (NSF Award Number: 2153040)** Boulder, CO
Environmental Data Science Innovation & Impact Lab (ESIIL)
University of Colorado, Boulder

2019 – 2024 **Doctoral Research Fellow** Philadelphia, PA
Decision Sciences & MIS Department
Drexel University

Fellowships & Research Experience (continued)

2019 – 2021 **Research Assistant** Salt Lake City, UT
Adler Lab - Mathematics Department
University of Utah

2018 – 2019 **Undergraduate Research Assistant** Salt Lake City, UT
Research Experience for Undergraduates (REU)
University of Utah

2018 **Computer Scientist (Internship)** Hill AFB, UT
309th Software Engineering Group
United States Air Force

Teaching Experience

2019 – 2024 **Instructor** Philadelphia, PA
Decision Sciences & MIS Department
Drexel University

Course	Level	Quarter(s)	Tool(s)
BSAN 360: Programming for Data Analytics	U	Winter 2022	R
Ph.D. Programming Bootcamp	PhD	Summer 2021; Summer 2022	Python
MIS 200: Management Information Systems (Recitation Section)	U	Fall 2019; Fall 2020; Winter 2021	MS Access; Excel; HTML

**Undergraduate (U)*

2019 – 2024 **Teaching Assistant** Philadelphia, PA
Decision Sciences & MIS Department
Drexel University

Course	Level	Quarter(s)	Tool
BSAN 360: Programming for Data Analytics	U	Spring 2021	R
BSAN 601: Business Analytics for Managers	MS; MBA	Spring 2024	Excel
MIS 612: Aligning Information Systems & Business Strategies	EMBA; MBA	Fall 2023	-
MIS 625: Management of IT Operations	MBA	Fall 2023	-
OPM 200: Operations Management	U	Spring 2020; Fall 2021; Spring 2023	Excel
OPM 341: Supply Chain Management	U	Spring 2021; Spring 2022; Fall 2022	Excel
OPM 344: Revenue Management	U	Fall 2022	Excel
OPR 320: Linear Models for Decision Making	U	Summer 2020; Spring 2021	Excel
STAT 201: Intro to Business Statistics	U	Winter 2020; Spring 2020; Fall 2021; Summer 2022; Spring 2023; Winter 2024	Excel
STAT 202: Business Statistics II	U	Summer 2021; Spring 2023	Excel
STAT 205: Statistical Inference I	U	Spring 2020; Fall 2021	Excel
STAT 206: Statistical Inference II	U	Summer 2021	Excel
STAT 510: Intro to Statistics for Business Analytics	MBA	Summer 2023; Winter 2024	Excel
STAT 642: Data Mining for Business Analytics	MS; PhD	Winter 2023	R

**Undergraduate (U)*

2018 – 2019 **Mathematics & Computer Lab Assistant** Salt Lake City, UT
T. Benny Rushing Mathematics Student Center
University of Utah

Presentations

2025 **California 30x30 Partnership 2025 Summit** *San Diego, CA*
Talk: Metrics with Meaning: Assessing, Tracking, and Supporting 30x30 Biodiversity Conservation

2025 **NASA Biodiversity and Ecological Conservation Team Meeting** *Washington, DC.*
Poster: Leveraging NASA Data to Guide Biodiversity Conservation Investments with the Trust for Public Land

2024 **AGU Annual Meeting (AGU24)** *Washington, DC.*
Poster: Exploring innovation in biodiversity conservation decision-making through open science and generative AI

2024 **AAAI Conference on Artificial Intelligence (AAAI-24)** *Vancouver, BC, Canada.*
Poster: Decision-making for land conservation: A derivative-free optimization framework with nonlinear inputs

2023 **MIT Sloan Rising Scholars Conference** *Cambridge, MA (Virtual)*
Talk: Decision-making for land conservation: A derivative-free optimization framework with nonlinear inputs

2023 **INFORMS Annual Meeting** *Phoenix, AZ.*
Talk: Decision-making for land conservation: A derivative-free optimization framework with nonlinear inputs

2023 **SIAM Conference on Optimization (OP23)** *Seattle, WA.*
Talk: Reserve design in biodiversity conservation

2023 **NEDSI Annual Conference** *Washington, D.C.*
Talk: Optimal land conservation decisions for multiple species

2021 **INFORMS Annual Meeting** *Anaheim, CA. (Virtual)*
Talk: Regularized step directions in conjugate gradient minimization for machine learning

2021 **SIAM Conference on Optimization (OP21)** *Virtual.*
Talk: Conjugate gradient methods for machine learning

2020 **INFORMS Annual Meeting** *Virtual.*
Talk: Efficient solution of portfolio optimization problems via dimension reduction and sparsification

Awards & Grants

2023 **Rising Scholar**
MIT Sloan School of Management

2023 **Graduate Student Travel Subsidy Award**
Drexel University

2023 **DEI & Environment and Sustainability Innovation Micro-Grant**
Drexel University

2023 **Teck-Kah Lim Graduate Student Travel Subsidy Award**
Drexel University

Awards & Grants (continued)

2023 **Student Travel Award**
Society for Industrial and Applied Mathematics (SIAM)

2022 **Teaching Assistant Excellence Award**
Drexel University

2021 **Teaching Assistant Excellence Award (Highly Commended)**
Drexel University

2021 **Student Travel Award**
Society for Industrial and Applied Mathematics (SIAM)

2019 **Undergraduate Research Scholar Designation**
University of Utah

Service

2025 **Panelist** *CU Boulder, Career Services*
Event: Femme in STEM

2024-Present **CIRES Science Pathways Speaker** *Cooperative Institute for Research in Environmental Science*
- Promoting science engagement by delivering flash talks at Colorado institutions.

2023 **Session Chair** *INFORMS Annual Meeting*
Session: Nonlinear Optimization in Machine Learning.

2023 **Session Organizer** *SIAM Conference on Optimization*
Session: Nonlinear Optimization and Applications.

2023 **Session Chair** *NEDSI Annual Conference*
Session: Land, Sand, and Plastic Management.

2022 **Panelist** *Drexel University*
Event: Graduate Teaching Assistance Orientation.

2019 **Mathematics Tutor (Volunteer)** *Utah Prison Education Project*
- Provided weekly tutoring sessions at the Utah State Prison.
- Supported students who are incarcerated and taking a Salt Lake Community College math course.